

This assignment will be undertaken in teams of 4 students. The teams will be self-selecting in the first instance and then randomly assigned as necessary. You will work on this assignment continuously during the practical sessions as a group.

Brief:

The United Nations has identified 17 Sustainable Development Goals. Included in this is the goal of 'Reduced Inequalities'. In this assignment you are required to design a novel technology that promotes **wellbeing** and **inclusivity** for a specified **underrepresented or vulnerable user group** (e.g., older adults, neurodiverse users, low-income communities) in line with this sustainable development goal. You may adopt design strategies such as participatory design, co-creation and co-design, service design, design for social innovation, inclusive design and open innovation. You may adopt an established technology or you may find opportunity in contemporary developments in technology, such as generative AI, digital fabrication, citizen sensing, the maker movement, the sharing economy, big data, social networks, IoT, gamification, new sensors and actuators, and Augmented/ Virtual Reality, to name just a few. Remember, though, that sometimes the best design solution or approach may flow from simple yet sharp insights uncovered from research, and might require only minimal technology – what is important is that your solution should be appropriate for the particular goal you are focusing on.

In addressing this design challenge, the following criteria should be considered as guidelines:

- Is your design clearly linked to the UN goal of Reduced Inequality?
- Does your design specify and solve a relevant and “burning” problem?
- Does your design use technology in an appropriate and novel way?
- Was the design well-crafted and effectively presented?
- Was the design validated in an appropriate and valid way to demonstrate the fulfillment of your design goal?
- Was relevant prior work properly identified and cited?

- Were analysis, synthesis, design, and evaluation systematic and sufficient?
- Was the design developed far enough to demonstrate the key ideas?
- Did the team explore the entire ecosystem of stakeholders, conditions, and contexts?

Submitting

1. **A paper:**
 - An introduction describing the overall idea and how it addresses the design challenge.
 - Evidence that you considered the stakeholders, conditions, and contexts in which your proposed solution would be used. E.g., a description of the important characteristics of the users of the system, the key tasks performed by users, and important characteristics of the task environment.
 - A description of the requirements and goals of your system. It is important this you demonstrate how these requirements and goals respond to both the design challenge and to the specific users, tasks and contexts identified.
 - A review of relevant HCI literature. This does not need to be extensive but should show how your solution relates on prior HCI literature.
 - Evidence of the initial ideation process where different options were considered. The key point here is to demonstrate how the requirements were used to develop a set of design alternatives for your problem. It is to come up with several different design ideas/concepts, not just a small set of variations from some basic design. The paper should include at least three different design concepts.
 - A summary of the design principles, approaches, and sources of inspiration applied in developing your solution.
 - A description of your proposed solution, the real educational problems that you are solving, and your main claims for your proposed solution. Teams may present either a concept (a clear, detailed design specification that can be expressed through a low fidelity prototype), or a fully realised prototype.

Either way, teams must clearly illustrate their design decisions and demonstrate the design processes that have been followed.

- You are not required to evaluate your solution with real users, but should apply other evaluation methods to validate your design.
- Any other supplementary materials you develop, e.g. personas, sketches, should be included in as an appendix to the paper. It is not included in the 8–page paper limit.

2. A Prototype

The paper should describe a final solution. Separate to this you must submit a prototype. This could take various forms, e.g., sketches, a presentation, a physical or software prototype. Using these materials you must then create a 5 minute video. This video should be submitted alongside the paper.

The video may illustrate how your solution fits the lives of the users with the help of scenarios, or addresses human aspects of the chosen sustainable development goals. It may also illustrate some details of the interface and the information presented. The Video may include:

- Examples of significant contextual data and its analysis (primary, secondary research, or both)
- Key creative sources of design inspiration (existing designs and systems)
- Sketches of the evolving solution
- Scenarios depicting how the solution fits in the life of users and solves problems / engages or entertains users
- Details of the interface and information design where relevant

3. An individual reflection

You must also submit a one–page personal reflection. This should reflect on the lessons learned in the assignment and describe your individual role in the team assignment.